

Mock Heat Flow Study

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Input	examples/dip/CreateReport/DIPfile

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1 Introduction

1.1 Purpose

This document describes a mock heat-flow experiment. The parameter tables show the **evaluated** values used for one run. All values, people, and publication references are illustrative.

1.2 Override example

The declared `experiment.flow_speed` is 2 m/s in `parameters.dip`. The project applies `overrides.dip`, which replaces it with 3 m/s. Its parameter block in the Parameters section shows the effective value, the original declaration, and the override source.

1.3 How to read the report

The sensor and analysis settings come from reusable schemas. The source and custom-unit sections explain where the other inputs originated.

1.4 Scope

The example includes two schemas, two custom units, multiple source files, publication details, and values with physical units. The linked contents page leads to each main report section.

2 Parameter guide

A quick index to the full reference entries. Scalars show effective values; arrays show shape and units.

Parameter	Value / summary
<code>experiment.analysis.method</code>	"mean"
<code>experiment.analysis.window</code>	8
<code>experiment.duration</code>	12 h
<code>experiment.flow_distance</code>	129.6 km
<code>experiment.flow_speed</code>	3 m*s-1
<code>experiment.inlet_temperature</code>	295 K
<code>experiment.interval</code>	1 sample_period
<code>experiment.outlet_temperature</code>	301 K
<code>experiment.readings.temperature</code>	Array (3) K
<code>experiment.readings.time</code>	Array (3) s
<code>experiment.sample_count</code>	60
<code>experiment.sensor.accuracy</code>	0.2 K
<code>experiment.sensor.model</code>	"T100"
<code>experiment.title</code>	"Heat & Flow_demo"
<code>experiment.track_length</code>	5 lab_length
<code>lab.operator</code>	"Example Research Team"
<code>lab.pressure</code>	1.013e5 Pa
<code>lab.room_temperature</code>	293 K

Inputs below are values read during evaluation; a read does not always mean the value was necessary for the result.

3 Parameters

<code>experiment.analysis.method</code>	
Value	"mean"
Type	str
Description	Method used to summarize the mock sensor readings
Applied schemas	analysis

Supplied by schema	analysis
Declared at	analysis.dipl:6
Declaration	method str = "mean"
experiment.analysis.window	
Value	8
Type	int64
Description	Number of readings in a summary window
Applied schemas	analysis
Supplied by schema	analysis
Declared at	analysis.dipl:8
Declaration	window int = 8
experiment.duration	
Value	12
Type	float64
Units	h
Description	Length of one run
Used by	experiment.flow_distance
Declared at	parameters.dip:8
Declaration	duration float = 12 h
experiment.flow_distance	
Value	129.6
Type	float64
Units	km
Description	Distance implied by effective flow speed and run duration
Expression	{?experiment.flow_speed} * {?experiment.duration}
Reads during evaluation	?experiment.flow_speed, ?experiment.duration
Declared at	parameters.dip:18
Declaration	flow_distance float = ({?experiment.flow_speed} * {?experiment.duration}) km
experiment.flow_speed	
Value	3
Type	float64
Units	m*s-1
Description	Nominal inlet speed, replaced for this run
Used by	experiment.flow_distance
Declared at	parameters.dip:16
Declaration	flow_speed float = 2 m/s
Override at	overrides.dip:1
Override	experiment.flow_speed = 3 m/s
experiment.inlet_temperature	
Value	295
Type	float64
Units	K
Description	Initial inlet temperature
Declared at	parameters.dip:20
Declaration	inlet_temperature float = 295 K
experiment.interval	
Value	1
Type	float64
Units	sample_period
Description	Time between sensor samples in the project's custom unit
Declared at	parameters.dip:12
Declaration	interval float = 1 sample_period
experiment.outlet_temperature	
Value	301
Type	float64

Units	K
Description	Initial outlet temperature
Declared at	parameters.dip:22
Declaration	outlet_temperature float = 301 K

experiment.readings.temperature	
Value	[295, 296, 297]
Type	float64
Shape	3
Units	K
Declared at	DIP0_FILE2_TABLE:2
Declaration	temperature float K

experiment.readings.time	
Value	[0, 1, 2]
Type	float64
Shape	3
Units	s
Declared at	DIP0_FILE2_TABLE:1
Declaration	time float s

experiment.sample_count	
Value	60
Type	int32
Description	Number of readings in the sample set
Declared at	parameters.dip:10
Declaration	sample_count int = 48
Modified at	environment.dip:9
Modification	experiment.sample_count = 60

experiment.sensor.accuracy	
Value	0.2
Type	float64
Units	K
Description	Stated temperature accuracy
Applied schemas	sensor
Supplied by schema	sensor
Declared at	sensor.dipl:9
Declaration	accuracy float = 0.2 K

experiment.sensor.model	
Value	"T100"
Type	str
Description	Sensor model identifier
Applied schemas	sensor
Supplied by schema	sensor
Declared at	sensor.dipl:7
Declaration	model str = "T100"

experiment.title	
Value	"Heat & Flow_demo"
Type	str
Description	A small mock experiment for the generated report.
Declared at	parameters.dip:2
Declaration	title str = "Heat & Flow_demo"
Authors	A. Example and B. Researcher
Publication	A demonstration of evaluated DIP reports
Year	2026
DOI	10.1234/example.heat-flow

experiment.track_length	
Value	5
Type	float64
Units	lab_length

Description	Length expressed in the custom project unit
Declared at	parameters.dip:14
Declaration	track_length float = 5 lab_length

lab.operator	
Value	"Example Research Team"
Type	str
Description	Team responsible for this mock run
Declared at	environment.dip:6
Declaration	operator str = "Example Research Team"

lab.pressure	
Value	1.013e5
Type	float64
Units	Pa
Description	Ambient pressure during the demonstration
Declared at	environment.dip:4
Declaration	pressure float = 101325 Pa

lab.room_temperature	
Value	293
Type	float64
Units	K
Description	Ambient laboratory temperature
Declared at	environment.dip:2
Declaration	room_temperature float = 293 K

4 Tables

Column values are listed under Parameters.

experiment.readings	
Rows	3
Columns	2
Column 1	time (float64, s)
Column 2	temperature (float64, K)

5 Schemas

Each schema lists the evaluated parameters it supplied. A schema may also apply to a path without supplying every value on that path.

analysis	
Description	Reusable analysis and reporting choices
Declared at	analysis.dip:6
Supplied parameters	experiment.analysis.method, experiment.analysis.window
Authors	C. Analyst
Publication	Illustrative data reduction method
Year	2026

sensor	
Description	Reusable temperature sensor settings
Declared at	sensor.dip:7
Supplied parameters	experiment.sensor.accuracy, experiment.sensor.model
Authors	A. Example and B. Researcher
Publication	A mock sensor calibration study
Year	2026
DOI	10.1234/example.sensor

6 Sources

DIP0	
Content hash	SHA-256: e3b0c44298fc1c14...
DIP0_FILE0	
Path	sensor.dipl
Parent	DIP0_project0
Parent line	9
Content hash	SHA-256: 3382c36674d1c213...
DIP0_FILE1	
Path	analysis.dipl
Parent	DIP0_project0
Parent line	9
Content hash	SHA-256: a80119fe87f05dcf...
DIP0_FILE2	
Path	parameters.dip
Parent	DIP0_project0
Parent line	12
Content hash	SHA-256: 904bb60ca7b3b325...
DIP0_FILE3	
Path	environment.dip
Parent	DIP0_project0
Parent line	12
Content hash	SHA-256: 42ca0d58a62f6701...
DIP0_OVERRIDE0	
Path	overrides.dip
Parent	DIP0_project0
Parent line	15
Content hash	SHA-256: 28aa6bc9d7497540...
DIP0_project0	
Path	DIPfile
Parent	DIP0
Content hash	SHA-256: af7ff59c1f9fd643...

7 Custom units

lab_length	
Definition	2*m
sample_period	
Definition	0.5*s